
Summary

Embedded software engineer with experience building and debugging software close to hardware. Comfortable working on Linux-based and constrained systems, with a strong focus on correctness and clear design. Used to taking ownership of real code in team settings and improving solutions through testing and iteration. Seeking roles focused on developing reliable embedded systems in real-world environments.

Education

MSc in Computer and Embedded Systems Engineering <i>Delft University of Technology</i> CGPA (2-year program): 7.5/10	Sept 2023 – June 2025 Delft, Netherlands
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B.E. in Electronics and Communication Engineering <i>Sir M Visvesvaraya Institute of Technology</i> CGPA (4-year program): 9.3/10	Aug 2018 – Aug 2022 Bangalore, India
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Skills and Certifications

Programming Languages	C, C++, Python, Rust
Microcontrollers	ATmega328P, ESP32, nRF51822 (ARM Cortex-M0)
Communication Protocols	SPI, I ² C, UART
Operating Systems	Linux
Tools & Frameworks	Git and Docker

Certifications: Industrial IoT on Google Cloud — Google; Programming for Everybody (Python) — University of Michigan

Experience

IMEC Graduate Research Assistant	Eindhoven, Netherlands Sep 2024 – July 2025
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- Analyzed state-of-the-art mapping tools (Timeloop, ZigZag) and identified their limitations in modeling event-driven accelerator architectures.
- Designed and developed AeDAM, a framework for design space and mapping exploration of AI workloads on event-driven architectures.
- Built analytical models to estimate energy, latency, and area for dense neural networks, and validated them through a case study on the SENECA neuromorphic architecture.
- Achieved up to $2.5\times$ faster exploration and up to 52% latency improvement over ZigZag, establishing AeDAM as an effective framework for event-driven accelerator optimization.

UVASKA Robotics Software Engineer	Bangalore, India Aug 2022 – Aug 2023
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- Led the development of the software stack for a custom 7-axis and 6-axis gantry-style articulated robot, including motion control, hardware interface integration, and system testing.
- Implemented kinematics and motion planning algorithms in C++ and Python, improving control precision.
- Evaluated and integrated modular software packages for system-level control and testing.
- Established structured version control workflows using Git, and collaborated within a 4-member team to iteratively test and debug software modules for a 6-axis industrial manipulator.

Epson Robotics Software Intern	Bangalore, India Mar 2021 – Sept 2021
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- Developed a Python-based interface on Raspberry Pi to control EPSON SCARA robots, bypassing proprietary SPEL-based software to simplify programming for new developers.
- Tested and validated the interface on physical robot hardware, improving accessibility and speeding up programming workflows within the R&D team.
- Assisted in a client project to build an automation system using the robot's vision module to detect specific shapes and perform sorting operations.

Projects

Embedded Quadcopter Control System

- Implemented a flight-control stack in Rust featuring a safety-first FSM, PID loops (roll/pitch/yaw), and Mahony sensor fusion at 200 Hz.
- Designed a fault-tolerant UART protocol with checksums and recovery logic; integrated joystick input (girs) and GUI telemetry (egui).
- Validated closed-loop stability and communication reliability through hardware-in-the-loop testing on a Cortex-M0 platform.

NES Emulator

- Built a cycle-accurate 6502 CPU emulator with memory-mapped I/O, interrupts, and cartridge mappers (NROM, MMC1).
- Integrated the PPU using the tudelft-nes-ppu crate and verified correctness against automated test ROMs.
- Ensured reliable execution of classic games such as Super Mario Bros and The Legend of Zelda through testing and system-level debugging.

TrackIn: Real-Time Indoor Tracking and Activity Recognition

- Built a smartphone-based tracking system integrating Bayesian localization, particle filtering, and an on-device TensorFlow Lite MLP classifier for real-time activity recognition.
- Processed multi-sensor data (Wi-Fi RSSI, accelerometer, gyroscope, magnetometer) using a 5th-order Butterworth filter and custom motion model, achieving 98% activity-classification accuracy.
- Deployed a compact Tiny-AI pipeline on Android with sub-second inference latency and validated localization consistency across 14 test cells.

Vargi Bots – Automated Warehouse Management System

- Designed a ROS-based warehouse automation system integrating dual UR5 manipulators, computer vision (OpenCV), and MQTT communication for autonomous warehouse operations.
- Developed modular Python APIs for robot coordination, real-time image processing, and priority-based task scheduling to optimize operational throughput.
- Simulated and validated pick-and-place workflows in Gazebo, improving system reliability through iterative debugging and ROS topic introspection.

Achievements

- **Best Idea Award (2018):** Led APEX (smart agriculture automation) to 1st prize and secured €1,000 funding to develop the solution.
- **Gov-Tec-Thon (2020):** Placed 15th/100 nationally, demonstrating scalability and societal impact of the project.
- **Eyantra Competition (2020):** Advanced to Round 2; ranked top 250/500 in IIT Bombay's robotics contest.

Publication

Artificial intelligence enabled plant emotion xpresser in the development hydroponics system 2021
Materials Today: Proceedings

Extracurricular activities

CoderDojo Delft, Netherlands

Mentor 2025 – Present

- Mentored children aged 8–12 in foundational programming concepts through hands-on, project-based sessions.
- Guided learners in problem solving, logical thinking, and translating ideas into simple working programs.

ROS Tutorials Channel Online

Content Creator 2021 – Present

- Created and maintain a YouTube channel with tutorials on the Robot Operating System (ROS), demonstrating the hardware implementation and communication frameworks.
- Produced practical, project-based videos to support learners and showcase applied knowledge in robotics development.

Languages & Interests

Languages: English, Hindi, Telugu

Interests: Reading non-fiction books, writing blogs, and playing badminton